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Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738062570685-Chapter 5 - unit 2 - Bringing Ideas to Life.pdf

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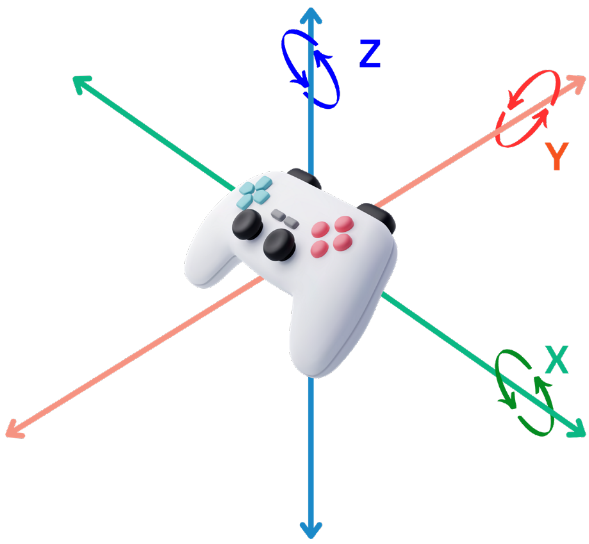
UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

Imagine you're the captain of a ship in an adventurous video game, steering through treacherous waters by tilting your device. This tilting action, whether forward, backward, or side to side, is similar to the x, y, and z angles we learned about.

Suppose you want your ship to race ahead, you tilt your device forward. The game detects this tilt using the x angle, propelling your ship forward. Tilting it back slows your ship down, as the game interprets this as a negative x angle.



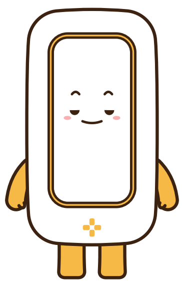
Now, imagine navigating your ship through narrow passages. Tilting your device to the left steers your ship left, thanks to the y angle. Tilting it right does the opposite, guiding your ship through the right path.

For sharp turns or dodging obstacles, you might twist your device. This action is picked up by the z angle. A twist to the left makes a swift left turn, and a twist to the right sends your ship darting right.

In our "MODI Compass System" project, we'll use an IMU module, similar to the technology in your game, to understand movement and direction. Just like in the game, the IMU module will help our creation sense which way it's facing, making our compass respond to the world around us, guiding us on our journey just like ancient travelers on the Silk Road!

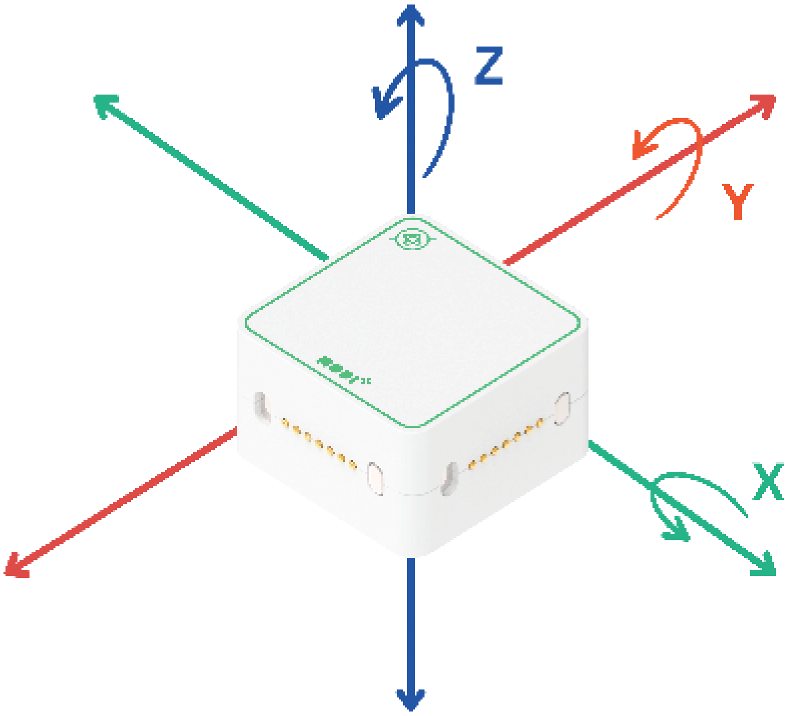
TIPS

IMU modules are like the secret agents inside our devices, helping them understand how we're moving and making our interactions with technology feel more intuitive and fun!



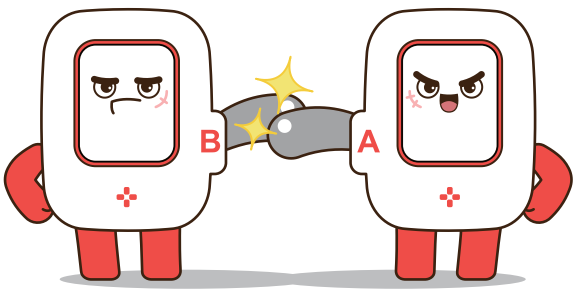
MODI COMPASS SYSTEM (BASIC)

Our MODI Compass System keeps a close eye on how much the IMU module tilts using something called a "while" loop in its program. When the IMU tilts a lot (more than 10 degrees or less than -10 degrees), the screen shows a tired face, meaning it's not quite balanced.



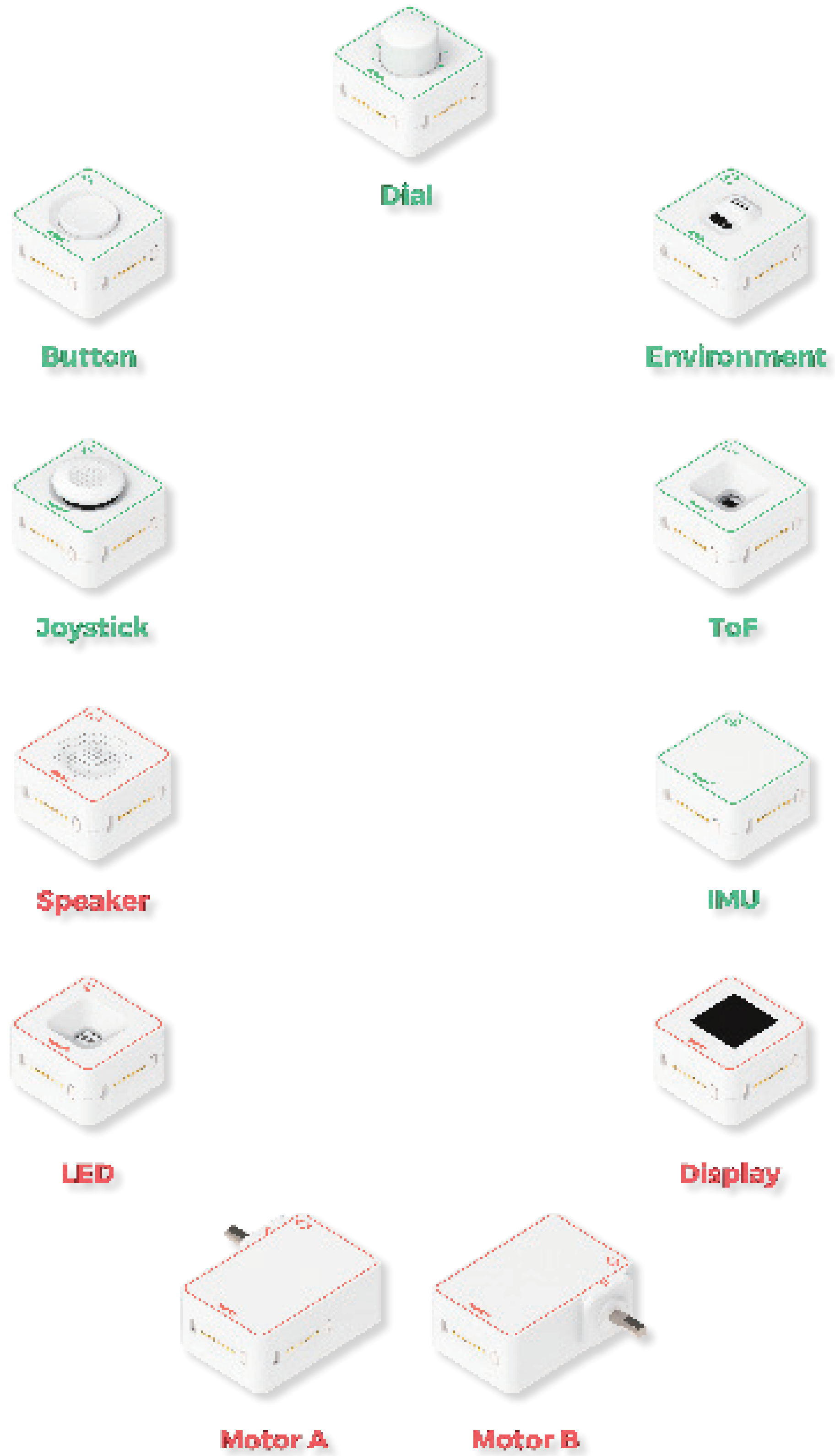
But, if the IMU stays almost flat, with a tilt between -10 and 10 degrees, you'll see a happy, smiling face on the screen. This part of our project uses just the Z angle of the IMU to show us how a compass works, teaching us about staying balanced and finding our way, much like ancient explorers did.

From now on, let's create a digital compass system with MODI modules that shows the correct direction to head!

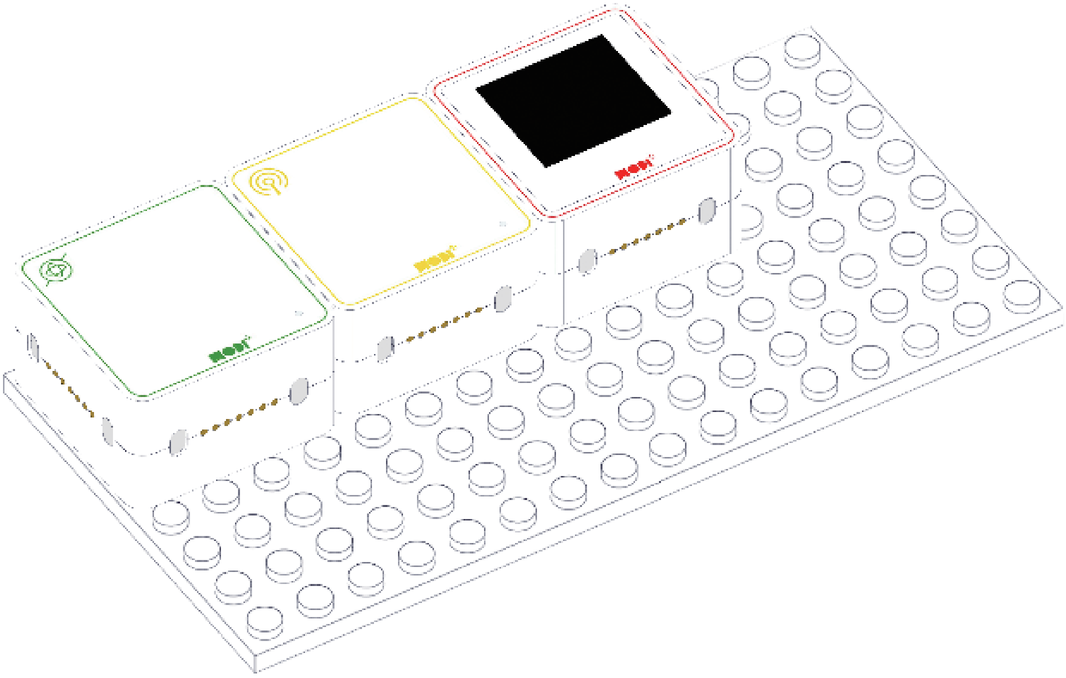


Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?

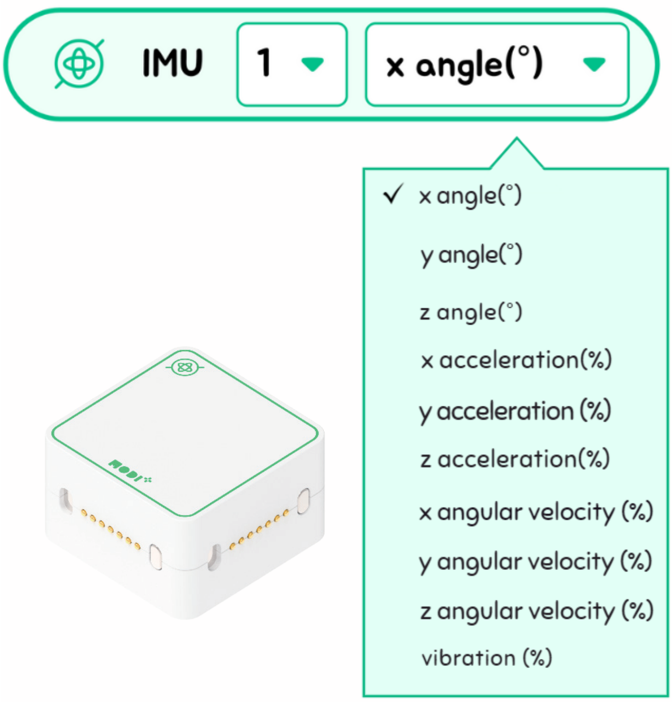


HOW DOES THE CREATION LOOK LIKE?



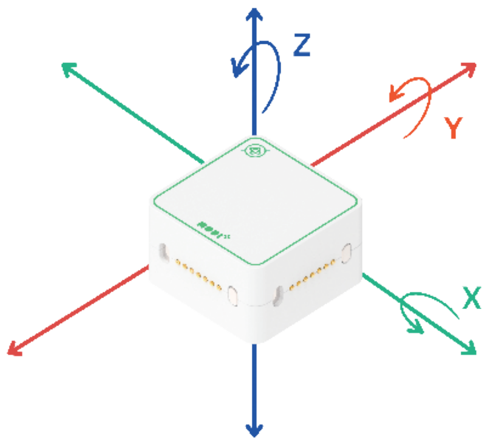
Module Functionality

Let's check the details of the input module below for this project



IMU

- **X, Y, Z Angle**: Shows how much the module is tilted.
- **X, Y, Z Acceleration**: Tells how fast the module moves in any direction.
- **X, Y, Z angular velocity**: Shows how quickly the module spins around.
- **Vibration**: Detects any shaking or vibrating motion.



1. X, Y, Z Angle	2 .Vibration:
Rotation angle measurements around each axis. Range: -180° ~ 180°	To measure it, there are 3 important parameters — acceleration, velocity and displacement. Range: 0% ~ 100%

TIPS

IMU can recevie
4 types of inpu data